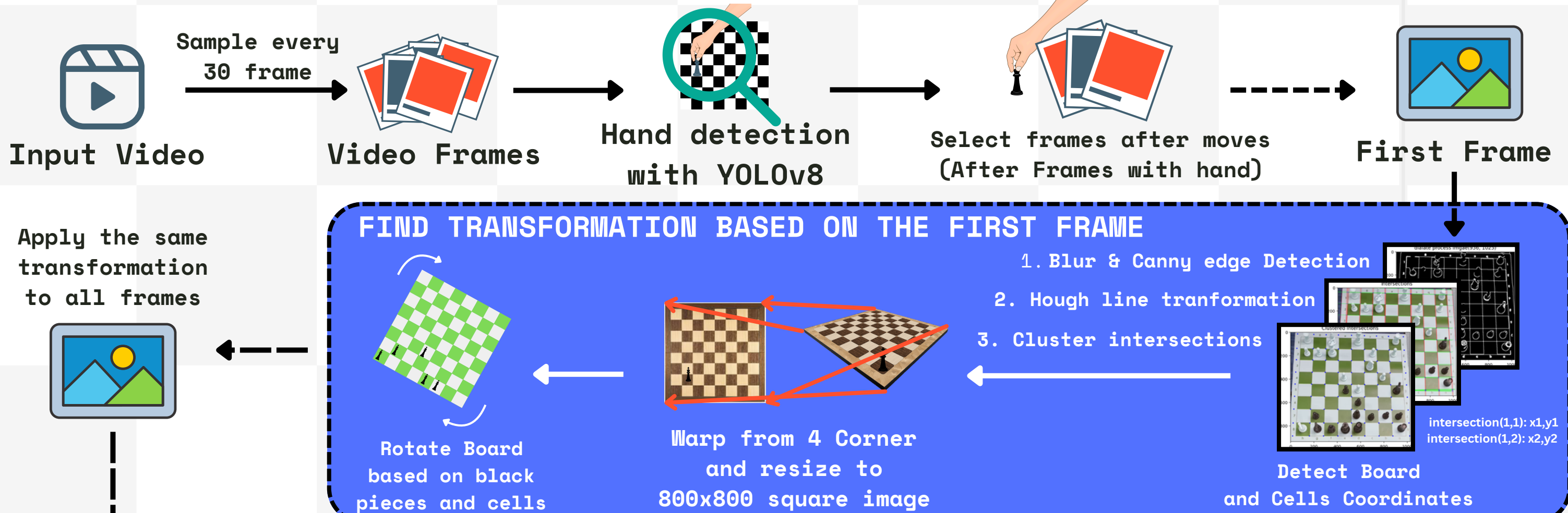


CHES DETECTION

BY 4D CHESS ENJOYER

PRE-PROCESSING

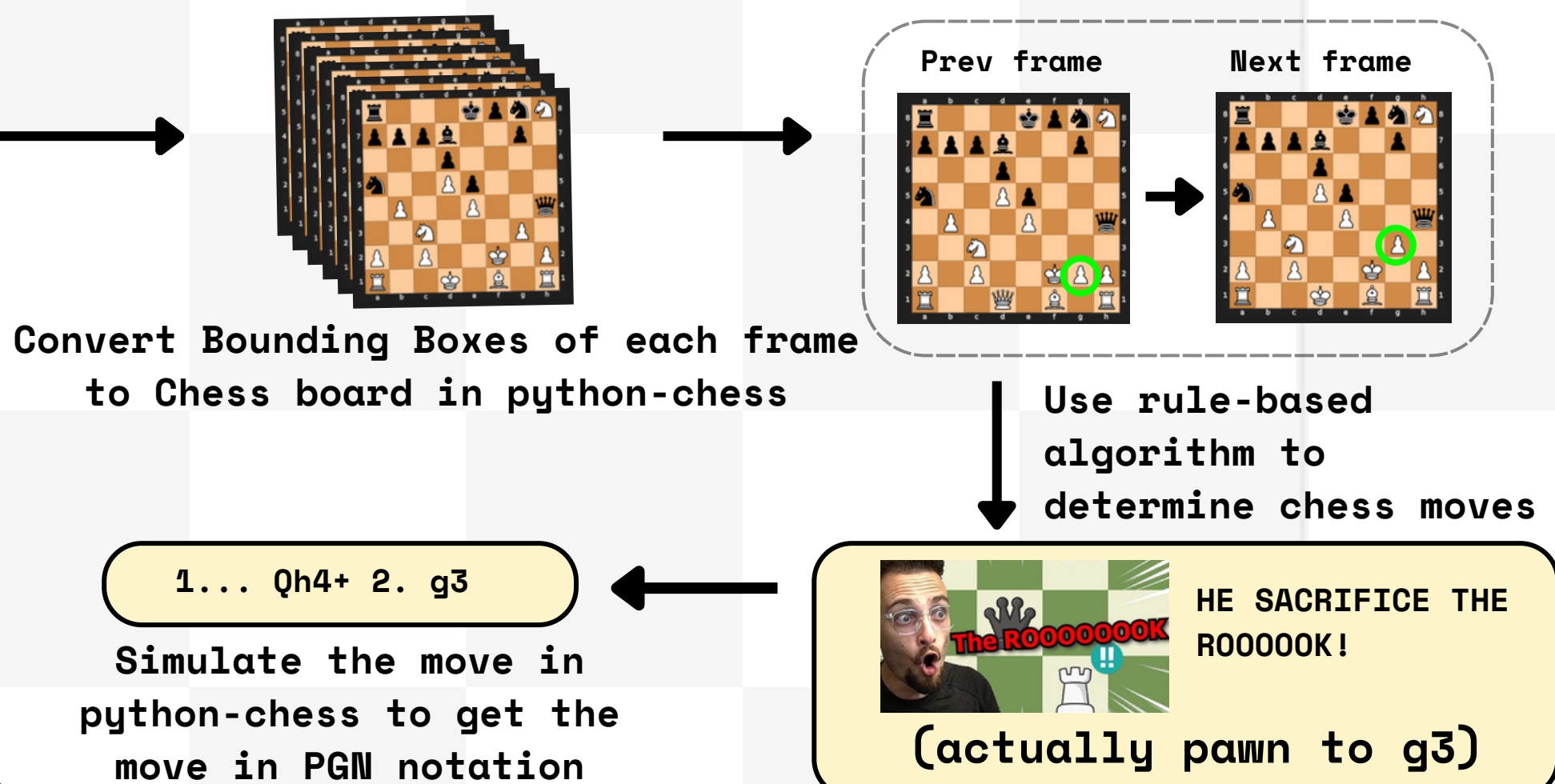


OBJECT DETECTION



Detect Chess Pieces with YOLOv8

POST-PROCESSING & PROBLEM SOLVING



MEMBERS & ROLES

Army (6532147621)

Pre-process (Cell coordinates detection) & Pipeline design & Model Training

Faith (6531330921)

Post-process & Research (Evaluation & Analysis)

Jeans (6531347621)

Pieces Detection Model & Data Engineer (Data Cleansing & Augmentation)

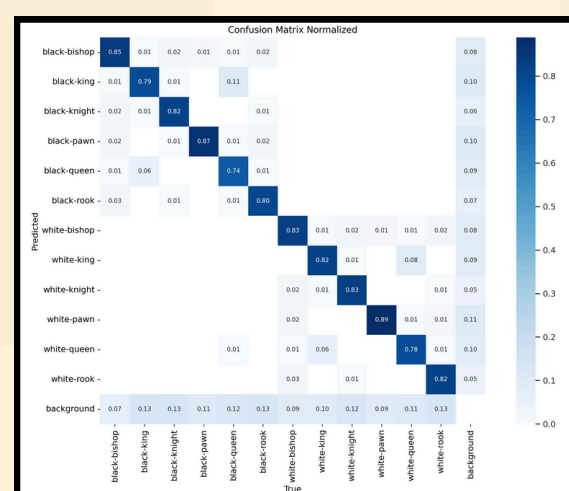
Junior (6531333821)

Post-process (Convert bboxes to Chess board & Solve problem with rule-based approaches) & Dataset Research

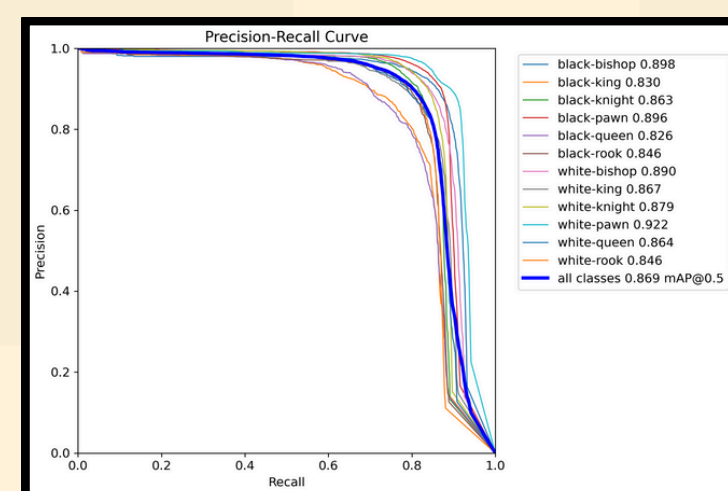
Paopeaw (6532149921)

Pre-process (Board rotate algorithm) & Hand Detection Model Research

EVALUATION METRICS & ANALYSIS



Confusion Matrix



PR curve

#	Team	Score
6	4D Chess Enjoyer	0.76

Score in Kaggle (0.76)

Higher Recall
Lower Precision

Pros

- Works well with slightly different chess piece designs and lighting
- Can detect illegal moves
- High recall piece detection
- Precise piece position detection
- Good at detecting chess piece movement

Cons

- Does not work well with different board designs and camera angles
- Require the board's corners to be visible in the data
- Bad if start at late-game
- Bad at detecting chess type

Evaluation Analysis

- Kings are often be confused as Queens
- Most pieces sometimes are detected as Bishops
- The model is good at differentiate between Black and White pieces
- With high chances of misdetection, The rule-based system may not detect moves or detect moves as wrong piece types.